$$J_{GRPO}(\theta) = \underbrace{\frac{1}{G}\sum_{i=1}^{G}\frac{1}{|a_i|}\sum_{t=1}^{|a_i|}}\{\min[\frac{\pi_\theta(a_{i,t}|s,a_{i,<t})}{\pi_{old}(a_{i,t}|s,a_{i,<t})}\hat{A}_{i,t}, clip(\frac{\pi_\theta(a_{i,t}|s,a_{i,<t})}{\pi_{old}(a_{i,t}|s,a_{i,<t})}, 1-\varepsilon, 1+\varepsilon)\hat{A}_{i,t}] - \beta D_{KL}[\pi_\theta||\pi_{ref}]\}$$

$$where\ \hat{A}_{i,t} = \frac{r_i - mean(r_1, r_2, \dots, r_G)}{std(r_1, r_2, \dots, r_G)}$$

Suppose the agent generates R rollouts, a rollout ρ produces N_ρ samples, the j_{th} sample has $|a_{\rho,j}|$ response tokens.

When treating R as the group size G:

$$J_{GRPO}^{rollout-mean}(\theta) = \frac{1}{R}\sum_{\rho=1}^{R}\frac{1}{\sum_{j=1}^{N_\rho}|a_{\rho,j}|}\sum_{j=1}^{N_\rho}\sum_{t=1}^{|a_{\rho,j}|}\{\dots\}$$

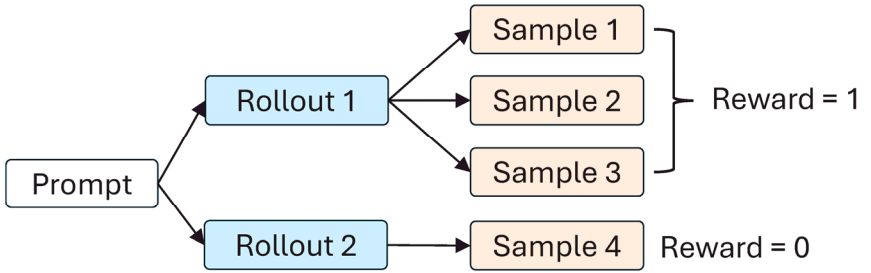
When treating $\sum_{\rho=1}^{R} N_\rho$ as the group size G:

$$J_{GRPO}^{sample-mean}(\theta) = \frac{1}{\sum_{\rho=1}^{R} N_\rho}\sum_{\rho=1}^{R}\sum_{j=1}^{N_\rho}\frac{1}{|a_{\rho,j}|}\sum_{t=1}^{|a_{\rho,j}|}\{\dots\}$$

When treating $\sum_{\rho=1}^{R}\sum_{j=1}^{N_\rho}|a_{\rho,j}|$ as the group size G:

$$J_{GRPO}^{token-mean}(\theta) = \frac{1}{\sum_{\rho=1}^{R}\sum_{j=1}^{N_\rho}|a_{\rho,j}|}\sum_{\rho=1}^{R}\sum_{j=1}^{N_\rho}\sum_{t=1}^{|a_{\rho,j}|}\frac{1}{1}\sum_{k=t}^{t}\{\dots\}$$

finer granularity with respect to group elements



When group mean and std is computed at rollout level:

$$\hat{A}_{i,t} = \frac{r_i - mean(r^1_{rollout}, r^2_{rollout})}{std(r^1_{rollout}, r^2_{rollout})} = \frac{r_i - mean(1,0)}{std(1,0)}$$

When group mean and std is computed at sample level:

$$\hat{A}_{i,t} = \frac{r_i - mean(r^1_{sample}, r^2_{sample}, r^3_{sample}, r^4_{sample})}{std(r^1_{sample}, r^2_{sample}, r^3_{sample}, r^4_{sample})} = \frac{r_i - mean(1,1,1,0)}{std(1,1,10)}$$